

Physics: Motion and Forces

Distance is the total length traveled, while displacement is the change in position in a particular direction. Average speed equals distance divided by elapsed time. Average velocity equals displacement divided by elapsed time.

Acceleration describes how quickly velocity changes. For constant acceleration, final velocity equals initial velocity plus acceleration multiplied by time: $v = u + at$.

Newton's first law says an object keeps its state of rest or uniform straight-line motion unless a net external force acts on it. Newton's second law is $F = ma$, where force is measured in newtons, mass in kilograms, and acceleration in metres per second squared.

Weight is the gravitational force on an object and can be calculated as $W = mg$ near Earth's surface. Mass measures inertia and does not change with location; weight can change when gravitational field strength changes.